Ibrar ul Hassan Akhtar, PhD | Curriculum Vitae

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Profile Summary

I started my career as a research associate in various public sector organizations in Pakistan. Thereafter entered private sector in the domain of marketing, sales and technical services related to agriculture sector. In 2006, I joined field of space science and technology research/applications and worked on various national and international projects at National Space Agency of Pakistan (SUPARCO). Notably technology development including Agriculture, Forestry, Water Resource Management, Atmospheric Science, Environmental Science, Food Security, Mountains Ecology and various others sectors of National development. Additionally, worked in collaboration with experts from FAO-UN Italy, Morocco, Hungary, France and USA. Currently Leading the "Technology Application in Natural Resource Management Group" at SUPARCO and developing system to address national food and water security and its policy making. Future Plans are to carry out research on "Integration of AI, EO and Science for the benefit of Humanity".

I am seeking an International career for future growth and contribution to Multi-disciplinary Research and Technology.

Research and Academic Interests

Remote Sensing and GIS Applications
Digital Image Processing, Geo-spatial Analysis, Optical, SAR, Hyper-Spectral, Hyper-temporal, Lidar Applications in agriculture, hydrology, forestry, urban modeling, soil and environment. Most common used satellite data include SPOT6/7, Pleiades, Pleiades-Neo, Superview, Sentinel1, Sentinel2, Sentinel5P, Landsat8, Landsat9 and others.

Climate and Meteorology
Agricultural Meteorology, Hydro-Meteorology, Climate Modeling, Climate Change Attributions, Impacts on agriculture and water resources, Urban climates, Drought mapping, Soil moisture analysis and other.

Agriculture and Forestry
Agricultural crops area and yield mapping, forest management, forest mapping including phenology, density and change analysis using ground and satellite data sets.

AI and Data Science
AI and Data science use in agriculture, forestry, water resources, urban and land us land cover modeling.

Environmental Modeling
Satellite and ground integrated mapping and monitoring of major air, soil and water pollutants.

Water Resources
Irrigation, rivers and basin scales water resources modeling and management

Disaster Management
Satellite and hydro-meteorological modeling based mapping of floods, droughts, landslides, cyclones and extreme events.

Work Experience

October 2006 – Present
Director/Research Scientist Space Applications and Research (Remote Sensing, Agriculture, Forestry, Water and Climatology), SUPARCO

1. **Research & Development**
   - ✓ Lead and oversee research projects in remote sensing applications for agriculture, forestry, water resources, and climatology.
   - ✓ Develop and implement innovative geospatial solutions for monitoring environmental and agricultural changes.
   - ✓ Conduct advanced research on climate change impacts using satellite and GIS technologies.
2. **Remote Sensing & GIS Applications**
   - ✓ Manage and analyze satellite imagery for precision agriculture, deforestation monitoring, water resource management, and climate studies.
   - ✓ Develop predictive models for drought, flood risk assessment, and crop yield forecasting using AI and remote sensing techniques.
   - ✓ Integrate satellite-based remote sensing data with ground-based observations for enhanced decision-making.
3. **AI & Machine Learning in Remote Sensing**
   - ✓ Develop and implement AI-driven models for satellite image analysis, classification, and predictive analytics.
   - ✓ Utilize deep learning and computer vision techniques for object detection, land cover mapping, and anomaly detection.
   - ✓ Improve data fusion techniques by integrating AI with remote sensing, GIS, and IoT sensor networks.
4. **Research & Development in Earth Observation**
   - ✓ Lead R&D initiatives focused on AI applications in climate modeling, precision agriculture, and water resource management.
   - ✓ Develop satellite-based models for drought prediction, deforestation monitoring, and urban expansion analysis.
   - ✓ Conduct advanced research on climate change impacts using remote sensing and AI-driven simulations.
5. **Smart Agriculture & Forestry Management**
   - ✓ Develop AI-powered crop monitoring and yield prediction models using multispectral and hyperspectral imagery.
   - ✓ Use remote sensing for pest and disease detection in agricultural fields.
   - ✓ Implement geospatial AI techniques for forest cover assessment and illegal logging detection.
6. **Water Resources & Climate Monitoring**
   - ✓ Apply AI to hydrological modeling for flood forecasting, groundwater mapping, and water scarcity analysis.
   - ✓ Develop climate resilience strategies by analyzing long-term satellite datasets using AI-based climate models.
   - ✓ Integrate remote sensing with AI for real-time weather monitoring and disaster response planning.
7. **Strategic Planning & Policy Development**
   - ✓ Define strategic goals for space-based Earth observation programs in Pakistan.
   - ✓ Collaborate with government agencies, research institutions, and international organizations for policy formulation on climate resilience and sustainable land management.
   - ✓ Develop national frameworks for using satellite data in environmental monitoring and disaster management.
8. **Project Management & Leadership**
   - ✓ Lead multi-disciplinary teams of scientists, engineers, and GIS specialists for high-impact research projects.
   - ✓ Oversee project budgets, timelines, and deliverables to ensure efficiency and effectiveness.
   - ✓ Foster collaboration with universities, NGOs, and industry stakeholders for capacity building and knowledge sharing.
9. **Capacity Building & Training**

- ✓ Develop training programs and workshops on remote sensing applications in agriculture, forestry, and climatology.
- ✓ Mentor junior scientists and researchers to enhance expertise in satellite-based environmental monitoring.
- ✓ Promote the adoption of advanced geospatial technologies within SUPARCO and partner organizations.

10. **Data Processing, Management & Decision Support**
    - ✓ Oversee the collection, processing, and interpretation of massive geospatial datasets.
    - ✓ Develop AI-driven decision-support systems for policymakers and government agencies.
    - ✓ Ensure automation of geospatial analysis for faster and more accurate insights.
11. **National & International Collaboration**
    - ✓ Represent SUPARCO in global AI and remote sensing research forums.
    - ✓ Collaborate with international organizations on joint AI and remote sensing projects.
    - ✓ Secure research funding for AI-driven Earth observation initiatives from national and international agencies.
12. **Leadership & Capacity Building**
    - ✓ Lead multidisciplinary teams of AI specialists, remote sensing scientists, and GIS experts.
    - ✓ Develop training programs and workshops on AI applications in agriculture, forestry, and water resource management.
    - ✓ Guide junior scientists in cutting-edge AI and remote sensing research.

01 January 2005- 31 August 2005

Research Associate, Ministry of Food and Agriculture, Pakistan

Scenario analysis and management of information received from local agriculture extension and Research departments in the form of cropped area and opinion yield survey of sugarcane, rice and Maize crop to make recommendation to government to rationalize their economic decisions like Import/exports, price control etc. Supervision of running research project on sugar beet introduction in Pakistan and sugar productivity enhancement. Collection and data analysis of experimental field data from different research location in Pakistan.

Educational Profile

2016-2021

Ph.D. Meteorology (Climatology/Remote Sensing with focus on water, food and climate) under Government of Pakistan Scholarship

Course work completed with >90% marks and 3.97/4.00 CGPA

Thesis Title "**Climate Integrated Remote Sensing Based Assessment of Agro-Ecosystem in Northern and Central Pakistan"**

COMSATS University Islamabad, Islamabad, Pakistan

2009-2010

MSc in Geomatic Engineering (**Environmental Mapping**)

Under HRD 2008-09 Scholarship

University College London, UK

2002-2004

M.Sc (Hon's) Agriculture

Top Position with Gold Medalist

University of Arid Agriculture Rawalpindi, Pakistan

1997-2001

B.Sc. (Hon's) Agriculture

Distinction

University of Arid Agriculture Rawalpindi, Pakistan

Professional Trainings

Feb 2024
Coding for Biologist
One day training by School of Applied Bioscience, NUST, Pakistan

August 2022
AI for Everyone
Online 4 week training by DeepLearning.AI (Coursera)

August 2022
Evaluating Ecosystem Services with Remote Sensing
ARSET, NASA

April-May 2022
Mapping Crops and their Biophysical Characteristics with Polarimetric Synthetic Aperture Radar and Optical Remote Sensing
ARSET, NASA

September 2021
Introduction and Access to Global Air Quality Forecasting Data and Tools
ARSET, NASA

January 2014
Geo Network System and Open Sources GIS Technology
FAO and United State Department of Agriculture (USDA)

June 2013
Training Workshop under AgMIP Project
Washington State University, University of Agriculture Faisalabad

July 2007-Sept 2007
Crop Yield Forecasting (Remote Sensing/GIS, Statistical Analysis, Agronomy, Climatic Analysis)
University de Liege, Longsway, Belgium

Oct 2006-April 2007
Agricultural Statistics from Remote Sensing & GIS
SPOT Image and Purpan Institute, France & SUPARCO, Islamabad

Sept 2001- Oct 2001
Data Analyst Certificate (SPSS & TSP)
International Islamic University (IIUI), Islamabad

Communication skills

- Strong interpersonal and communication skills
- Professional, articulate and confident speaking abilities
- Proper planning and time management
- Flexible to adapt changes in accordance with the working conditions
- Experience of experimental research work
- Ability to work independently and as a team member to produce good quality results

Computer skills and Competences

- Microsoft Office (Word, Excel, PowerPoint, Access, Project)
- Data Analysis Software's (R-Console, SPSS, Statistica, Minitab, TSP, SAS and many other)
- Image Processing through MATLAB and IDL
- Geographical Software's (Arc-GIS, Erdas imagine, Objectland, Multispec, ENVI)
- Crop Modelling Software's (DSSAT, Aquacrop, CropWat, APSIM, AMS, VAST, LCCS, WinDISP)
- AI Tools

Research Papers/Opinion Articles

1. **Akhtar, I. H.** and A. Khaliq. 2003. Impact of Plant Phenology and Coccinellids Predators on the Population Dynamics of Rose Aphids *Macrosiphum rosaeiformis* Das on Rose. Asian Journal of Plant Sciences, 2(1): 119-122.
2. **Akhtar, I. H.** and M. Aslam. 2003. Antixenosis Test on Red Flour Beetle *Tribolium castaneum Herbst*. Against Different stored Product Commodities under Lab. Conditions. Pak. J. Biol. Sci., 6(13): 11761178.
3. Mustafa, T., H. Javed and **I. H. Akhtar.** 2003. Impact of Environmental Factors and Physio-morphic Characters of Sunflower (*Helianthus annus*) Genotypes on Insect Pest Complex. Asian J. Plant Sci., 2(1724): 1166-1169.
4. Article by **Akhtar, I. H.** 2003. Pest Management of Wheat in Rainfed area of Pakistan in newspaper "The News" (13th December, 2003).
5. **Akhtar, I. H.**, H. Javed and A. Khaliq. 2004. Microclimatic Morphs and Plant Distribution Analysis of Rhopalosiphum padi (Fitch) and Schizaphis graminum (Rondani) on wheat. Asian J. Plant Sci., 3(4): 516521.
6. **Akhtar, I. H.**, H. Javed and A. Shakoor. 2004. Microclimatic Morphs and Plant Distribution Analysis of Cereal Aphids on Wheat. Asian Journal of Plant Sciences 3 (5): 539-543.
7. **Akhtar, I. H.** 2004. Impact of Host Plant Resistance and Abiotic Factors on Cereal Aphids of Wheat. M.Sc. (Hon's) Thesis. University of Arid Agriculture, Rawalpindi, 101 p
8. **Akhtar, I. H.** 2004. Geographical Information System Implication for Agriculture Trend Estimation in Rawalpindi District at Tehsil Level. IGIS, NUST, 34 p.
9. **Akhtar, I. H.**, H. Javed and E. Haq. 2007. Microclimatic distribution analysis of Bird cherry oat Aphid, Rhopalosiphum padi L. on wheat. Pakistan Journal of Zoology. 39(1): 21-25.
10. **Akhtar, I. H**., M. Naeem, H. Javed and M. A. Aziz. 2009. Could Host Plant Resistance effect on Cereal Aphids (Aphididae: Homoptera) population on Wheat Crop. Int. J. Agr. Vet. Med. Sci., 3: 31-42
11. **Akhtar, I. H**., A. Ghafoor, S. Z. Bokhari and Obaid ur Rehman. 2012. Drought Mining of Spot Vegetation Temporal Data for Prevailing Food Insecurity in Potohar Region. Proceedings on ISNET/CSE workshop of Space Applications for Food Security, page 130-136, http://www.isnet.org.pk/downloadables/ISNET%20CSE%20Proceedings.pdf
12. **Akhtar, I. H.** 2012. Pakistan needs a new crops forecasting system. SciDev. http://www.scidev.net/global/climate-change/opinion/pakistan-needs-a-new-crop-forecasting-system.html

13. Ahmad, I., A. Ghafoor, M. I. Bhatti, I. H. Akhtar et. al., 2014. Satellite Remote Sensing and GIS based Crops Forecasting & Estimation System in Pakistan, FAO expert meeting February, 2014, LAO PDR, 16 Pp. http://www.fao.org/
14. **Akhtar, I. H.** 2015. Pakistan needs a new Disaster Mitigation Strategy. Sci Dev. http://www.scidev.net/global/climate-change/opinion/pakistan-needs-a-new-crop-forecasting-system.html
15. Bontemps, S., M. Arias, C. Cara, G. Dedieu, E. Guzzonato, O. Hagolle, J. Inglada, N. Matton, D. Morin, R. Popescu, T. Rabaute, M. Savinaud, G. S. Canto, S. Valero, I. Ahmad , A. Begué, W. Bingfang, D. de Abelleyra, A. Diarra , S. Dupuy, A. French, **I. H. Akhtar** , N. Kussul, V. Lebourgeois, M. Lepage, T. Newby, I. Savin, S. R. Verón , B. Koetz and P. Defourny. 2015. Building a Sentinel-2 like data set specifically dedicated to agriculture monitoring over 12 sites globally distributed. Remote Sensing. 7(12), 16062-16090; doi: 10.3390/rs71215815. (IF: 3.7)
16. Juan Laso Bayas, Myroslava Lesiv, Francois Waldner, Schucknecht Anne, Martina Duerauer, Linda See, Steffen Fritz, Dilek Fraisl, Inian Moorthy, Ian McCallum, Christoph Perger, Olha Danylo, Pierre Defourny, Javier Gallego, Sven Gilliams, **I. H. Akhtar**… and John Wilson. 2017. A global reference database of crowd sourced cropland data collected using the Geo-Wiki platform". Scientific Data 4, 170136. (IF: 7.2)
17. Myroslava Lesiv, Juan Carlos Laso Bayas, Linda See, Martina Duerauer, Domian Dahlia, Neal Durando, Rubul Hazarika, Parag Kumar Sahariah, Mar'yana Vakolyuk, Volodymyr Blyshchyk, Andrii Bilous, Ana Perez-Hoyos, Sarah Gengler, Reinhard Prestele, Svitlana Bilous, **I. H. Akhtar**,….. and Steffen Fritz. 2019. Estimating the Global Distribution of Field Size using Crowdsourcing. Global Change Biology, 25 (1): 174-186. DOI:10.1111/gcb.14492 (IF: 10.9).
18. **Akhtar, I. H.** and H. Athar. 2019. Co-variability of water reservoirs with precipitation and climatic oscillations in Pakistan. Scientific Reports, https://www.nature.com/articles/s41598-019-54872-x (IF: 4.4).
19. **Akhtar, I. H.** and H. Athar. 2020. Water supply and effective rainfall impacts on major crops across irrigated areas of Punjab, Pakistan. Theoretical and Applied Climatology, 143, 1097-116. (IF: 3.19).
20. **Akhtar, I. H.** and H. Athar. 2021. Climate variability of wheat yields across canal irrigated areas of Punjab, Pakistan. Theoretical and Applied Climatology, 143, 1195-1211. (IF: 3.4).
21. Myroslava Lesiv, Dmitry Schepaschenko […] , **I. H. Akhtar**, […] and Michael Oberstiener. 2021. Global forest management data for 2015 at a 100 m resolution. Scientific Data, 9, 199. https://www.nature.com/articles/s41597-022-01332-3 (IF: 8.50).
22. **Akhtar, I. H.** 2023. Exploring COVID-19 Initial Pandemic curve through Statistical Evaluation. Acta Scientifica Malaysia, 7, 8-16.
23. **Akhtar, I. H.** and A. Ali. 2024. Space Technology Applications and Air Pollution Monitoring. COMSTECH Journal of Science Communication and Diplomacy, 1(1), 15-31. https://drive.google.com/file/d/1Jxd8Yi8Zt637t0tgiaarFZVGICtNofZ_/view
24. **Akhtar, I. H.,** S. Rahman and S. Khan, "Satellite Land Cover Monitoring Development and Assessment of Plantation Activities in Khyber Pakhtunkhwa, Pakistan During 2007-2024," *2024 International Conference*

on Frontiers of Information Technology (FIT), Islamabad, Pakistan, 2024, pp. 1-6, doi: 10.1109/FIT63703.2024.10838454.

25. **Akhtar, I. H** and M. Mehboob, 2025. Leveraging Space Science and Technology for Climate Induced Disaster Management in Pakistan, CISS Insight Journal. https://journal.ciss.org.pk/index.php/ciss-insight/article/view/410/285
26. Ali, A., **Akhtar, I.H.**, Raza, M. and Ali, A., 2025. National-Scale Orchard Mapping and Yield Estimation in Pakistan Using Object-Based Random Forest and Multisource Satellite Imagery. *Sensors*, *25*(24), p.7468. (IF: 3.5).
27. Mushtaq, Z., I. Javed, H. Ejaz, and **Akhtar, I.H.,** 2025. Multispecies Connectivity Assessment and Habitat Suitability Modeling for Common Leopard and Musk Deer in Azad Jammu & Kashmir". Conservation Science and Practice, under review (IF 2.8).
28. **Akhtar, I. H**. 2025. Investigation of Simulated Weather Data for Diverse Climates of Pakistan Using Markov Chain and Artificial Neural Network Models. Discover Atmosphere, under review.
29. **H. Athar,** M. Shoaib, F. Saeed, **Akhtar, I. H**. 2025. Transmission of Malaria Intensity in Changing Climate of Pakistan. International Journal of Biometeorology. Under Review (IF: 3.2).

Technical Reports and Book Chapters:

30. Khan, M.H., I. Ahmad, M. I. Bhatti, M. Asif and **I. H. Akhtar**. 2007. International FAO Training in Crop Yield Forecasting Modelling at University de Liege, Belgium, http://www.hoefsloot.com/Downloads/pakistan_UTF-101-crop_forecasting_training.pdf 30 Pp.
31. Bussay, A and **I.H. Akhtar**. 2008. Wheat yield/production forecasting and estimation Technology. Pakistan Space and Upper Atmosphere Research Commission, Islamabad, Pakistan. 25 Pp.
32. Bussay, A and **I.H. Akhtar**. 2009. Crop yield/production forecasting and estimation Technology for Kharif Crops (Cotton, Rice & Sugarcane). Pakistan Space and Upper Atmosphere Research Commission, Islamabad, Pakistan. 62 Pp.

MS Thesis:

33. **Akhtar, I. H**. 2011. Crop Yield Forecast Modelling: A case study of Sugarcane in Pakistan from Combined Remote Sensing and Agro meteorological Techniques. VDM Verlag Dr. Müller, Germany, 112 pages.

PhD Thesis:

34. **Akhtar, I. H**. 2020. Climate Integrated Remote Sensing based Agro-ecosystem assessment in Northern and Central Pakistan. COMSATS University Islamabad, Pakistan, 153 pages.

Abstract:

35. Hanif, M., **I. H. Akhtar**, Obaid-ur-Rehman, S. Iqbal and M. Javid. 2011. Monitoring Flood 2010: Satellite based Rapid Crop Damage Assessment. National Conference on “Sustainable Agriculture in Changing Climate” July 7-9, 2011, Bara Gali, Pakistan, p-42.

36. Obaid ur Rehman, M. Hanif, **I. H. Akhtar**, S. Iqbal and M. Javid. 2011. RS-GIS based Crop Monitoring and Forecasting System. National Conference on "Sustainable Agriculture in Changing Climate" July 7-9, 2011, Bara Gali, Pakistan, p-74.
37. **Akhtar, I. H.** 2012. Dengue and Possibilities in Space Technologies. National Conference on Science, January 10-12, 2012, Rawalpindi, Pakistan, p-163.
38. **Akhtar, I. H** and Lewis, P. 2012. Sugarcane Yield Forecast Modelling based on Remote Sensing and Agro meteorological Data. National Conference on Science, January 10-12, 2012, Rawalpindi, Pakistan, p-167
39. **Akhtar, I. H** and M. J de Smith. 2012. Geospatial Prediction Models for Sulphur Dioxide and GIS Based Health Risk Assessment for London, UK. National Conference on Science, January 10-12, 2012, Rawalpindi, Pakistan, p-164.
40. **Akhtar, I. H**. 2013. Comparative Wheat Crop Growth Monitoring through Remote Sensing based Crops Intelligence System. **AgMIP – Pakistan Kickoff Workshop & International Seminar on Climate Change**, 4-6 June, 2013, Faisalabad, Pakistan, p-90.
41. **Akhtar. I. H** *et al.,* 2014. Multi satellite sensors based agro-ecological dynamics in mountainous area of Mansehra district, Khyber Pakhtunkhwa, Pakistan. Global Space Applications Conference, 4-6 June, 2014, Paris, France
42. **Akhtar. I. H** and A. Hussain. 2018. Trans-boundary hydro-ecological and Spatio-temporal characterization of large water reservoirs. United Nations/Pakistan/PSIPW 4th International Conference on Space Technology for Water Management, 26 February - 2 March 2018, Islamabad, Pakistan.
43. **Akhtar, I. H.** 2018. Satellite Data based Land Use Land Cover Mapping of Pakistan. Inception Workshop on Land Degradation Neutrality (LDN) under United Nation Convention to Combat Desertification (UNCCD) for Pakistan. Ministry of Climate Change, 7-8 August, 2018, Islamabad. Pakistan.
44. **Akhtar, I. H**. 2020. Geospatial Monitoring of Desert Locust in Pakistan under UNOOSA and IWMI webinar, 12 June, 2020.
45. Participated in UN/Austria Symposium on "Space Applications for Sustainable Development Goal 13: Climate Action during 1-3 September, 2020. Online
46. **Akhtar, I**. H. and M. Iqbal. 2022. Smart Agriculture in Pakistan: Technology Perspective. International Conference on Space 2022 on Role of Space Technology and Applications in Socio Economic Development, 28-30 March, 2022, Islamabad, Pakistan.
47. **Akhtar, I. H.** 2023. Crops Vulnerability against Floods in Pakistan. National Consultative Workshop on National Catastrophe Modeling by SUPARCO and NDRMF, 13 February, 2023, Islamabad, Pakistan.
48. **Akhtar, I. H.** 2023. Geospatial Assessment of Forest Cover Status under changing climate in Khyber Pakhtunkhwa during 2007-2018. International Conference on Remote Sensing, GIS and Climate Change" (13-15 March 2023), University of the Punjab, Lahore, Pakistan.

Training as Resource Person:

1. International training workshop on Space Technology Applications for Food Security. 9-14 July, 2012. Dakar, Senegal. Funded by Inter Islamic Network for Space Sciences and Technology (ISNET).

2. Advanced Training on Crops Area Estimation through Satellite Technology for Punjab Agriculture Department staff 12-16 November, 2012, Lahore, Punjab, Pakistan. (Funded by FAO of the UN under GCP/PAK/125/USA project).
3. Advanced Training on Crops Area Estimation and Yield Modelling through Satellite Technology for Punjab Agriculture Department staff 21-25 January, 2013, Quetta, Balochistan, Pakistan. (Funded under Public Sector Development Project by Government of Pakistan).
4. Advanced Training Crops Yield Forecast and Estimation System for Punjab Agriculture Department 8-12 April, 2013 at Lahore, Punjab, Pakistan. (Funded by FAO of the UN under GCP/PAK/125/USA project).
5. Advanced Training Crops Yield Forecast and Estimation System for Punjab Agriculture Department May, 2013 at Lahore, Punjab, Pakistan. (Funded under Public Sector Development Project by Government of Pakistan).
6. Agriculture Information System – Building Provincial Capacity for Crops Forecasting and Estimation - 10th -16th June 2013 at Islamabad, SUPARCO. (Funded by FAO of the UN under GCP/PAK/125/USA project).
7. Advanced Training on Crops Yield Forecast and Estimation System for Punjab Agriculture Department September, 2013 at Lahore, Punjab, Pakistan. (Funded under Public Sector Development Project by Government of Pakistan).
8. 3rd Online Advanced Training on Remote Sensing application for forest Land Use Land Cover and Thematic Change Detection under KP Forest project, 21-24 December, 2020.
9. 4th Online Advanced Training on Remote Sensing application for Forest Phenology and Density Mapping under KP Forest project, 5-9 April, 2021.
10. Advanced Training on Geospatial Monitoring of Crops during 6-10 September, 2021 at SPARC-Islamabad, Pakistan. (Funded under Public Sector Development Project by Government of Pakistan).
11. Upcoming 1st training course on "Satellite and Aerial Remote Sensing" 10-14 January, 2021 under Joint PSDP project of DDSDP with PMAS Arid Agriculture University, Rawalpindi

Achievements

1. Gold Medalist in MSc (Hons) Agriculture Sciences (2004)
2. SUPARCO HRD Scholarship (2008-09)
3. Best Performance Award by SUPARCO (2013)
4. Team Achievement Award by SUPARCO (2013 and 2014)
5. Certificate of Achievement for research contribution as "Citizen Scientist" by International Institute for Applied Systems Analysis (IIASA), Vienna, Switzerland. (2017)

List of References

1. Prof Dr. Athar Hussain
   Department of Meteorology
   Head, Centre for Climate Research and Development
   COMSATS University Islamabad, Pakistan
   Cell: +92 331-5033376
   Email: athar.hussain@comsats.edu.pk
   URL: islamabad.comsats.edu.pk

2. Dr. Jan Dempewolf
   Associate Research Professor,
   Department of Geographical Sciences
   University of Maryland, USA
   ++1 (301) 405-3074
   dempewol@umd.edu

3. Prof Dr. Muhammad Naeem
   Chairman, Department of Entomology
   PMAS Arid Agriculture University, Rawalpindi, Pakistan
   Office: +92 51-9290013
   Cell: +92 333-5751475
   Email: Muhammad.naeem@uaar.edu.pk
   URL: uaar.edu.pk

4. Dr. Muhammad Imran Shahzad
   Associate Professor and Head,
   Department of Meteorology
   COMSATS University Islamabad, Pakistan
   Cell: +92 336-5592366
   Email: imran.shahzad@comsats.edu.pk
   URL: islamabad.comsats.edu.pk